```
    --dbname rideshare_development \
    -c "$query" & # separate processes
done
```

If you've configured PostgreSQL as indicated above, the 4th query requires a database connection and requesting one produces an error because no connections are available. You'll see output like the following.

sh/exhaust_database_connections_output.sh

```
running query SELECT pg_sleep(30). times: 1
running query SELECT pg_sleep(30). times: 2
running query SELECT pg_sleep(30). times: 3
running query SELECT pg_sleep(30). times: 4
psql: error: connection to server on socket "/tmp/.s.PGSQL.5432" failed \
 FATAL:  sorry, too many clients already
```

Running out of database connections is a severe problem.

- In an application with increased use, more database connections will be needed
- Your organization may not to wish to scale up (vertically) the database server, which does add CPUs and makes it possible to serve more connections, but costs more money
- Your application may keep a large amount of idle connections open, using up connections
- Idle connections from a long running query use up connections

For a Ruby on Rails application, on the conservative side, queries should be finishing in less than 1 second. For an application in a scaling mode under heavy concurrency, queries should run in the tens of milliseconds.

Run these queries to explore the active connections:

sql/idle_transactions.sql

```
-- e.g.
-- count,state
-- 7,active
-- 3510,idle
SELECT count(*), state
FROM pg_stat_activity
GROUP BY 2;

---https://dba.stackexchange.com/a/39758
SELECT * FROM pg_stat_activity
WHERE (state = 'idle in transaction')
AND xact_start IS NOT NULL;
```

How many connections do you need? Conduct some capacity planning and resource utilization investigation.